

```

ESP32S2 Dev Module
WithRound.ino Free_Fonts.h
1 #include "SPI.h"
2 #include "TFT_eSPI.h" // Graphics and font library for ILI9341 driver chip
3 #include "Free_Fonts.h" // Include the header file attached to this sketch
4 #include <EmonLib.h> // Include EmonLib Library
5
6 #include <WiFi.h>
7 #include <ESPAsyncWebServer.h>
8 #include <AsyncTCP.h>
9
10 AsyncWebServer server(80);
11 AsyncWebsocket ws("/ws");
12
13 EnergyMonitor emon1; // Create an instance for energy monitoring
14
15 // Use hardware SPI
16 TFT_eSPI tft = TFT_eSPI();
17
18 TFT_eSprite sprite = TFT_eSprite(&tft); // Create sprite object "sprite"
19
20 int currentX = 30; // Start position adjusted to the left
21 int graphStartX = 30; // Start position of the graph adjusted to the left
22 int numberStartX = 15; // Start position of the numbers adjusted to the left
23
24 // Variables for energy calculation
25 double realPower = 0; // Variable to store real power (Watts)
26 double supplyVoltage = 230.0; // Assuming a 230V supply voltage (adjust according to your region)
27 double current = 0; // Variable to store current (mA)
28 double energy = 0; // Variable to store total energy usage (kWh)
29 unsigned long lastUpdateTime = 0;
30 unsigned long updateInterval = 1000; // Update interval in milliseconds
31 double unitRate = 0; // Unit rate (cost per kWh)
32 double totalBill = 0; // Variable to store total bill
33
34 int maxCurrent = 20; // Initial max current value for scaling
35 int maxPower = 100; // Initial max power value for scaling
36
37 const char index_html[] PROGMEM = R"rawliteral(
38 <!DOCTYPE HTML>
39 <html>
40 <head>
41 <title>Energy Monitor Dashboard</title>

```

Fig. 4: Source code snippet

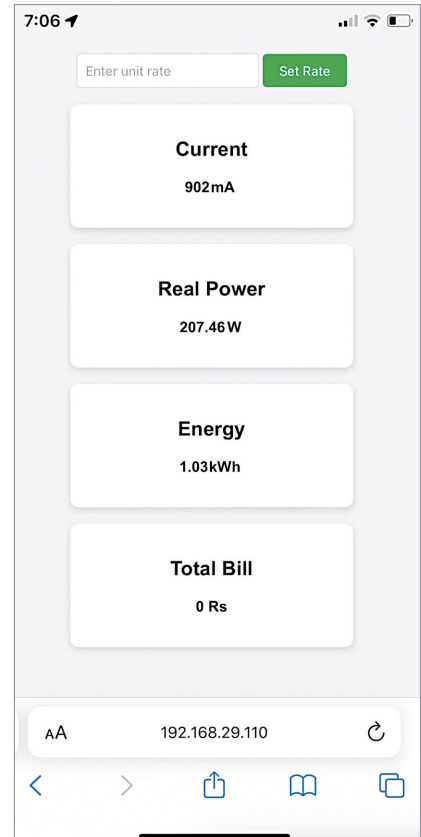


Fig. 6: The IoT meter's reading

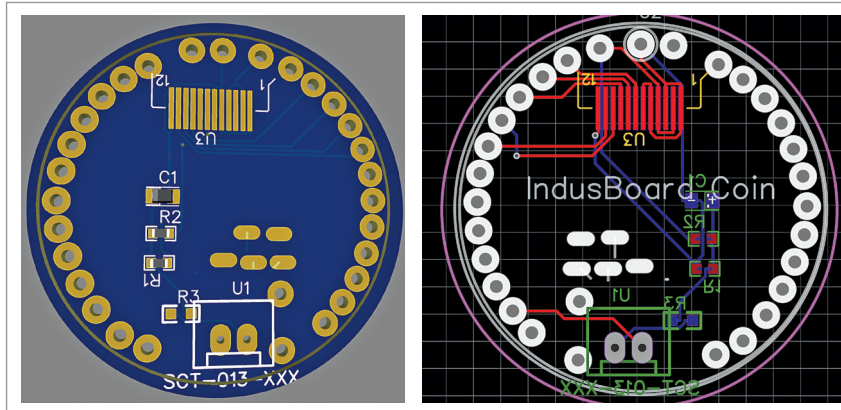


Fig. 5: PCB design of the IoT meter



Fig. 7: Final data of IoT meter on round display

meter in the Wi-Fi network list on any phone and connect to it. Open a browser and enter the device's internet protocol (IP) address to view the readings. Fig. 6 shows the data on the

OLED display, while Fig. 7 shows readings and a real-time graph on the round display. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/>

videos/slideshows/diy-iot-tracking-system-with-dashboard

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